

KHALIDH AHAMED

khalidh98@gmail.com | LinkedIn: www.linkedin.com/in/khalidh-ahamed | +917539903768

About

Software Engineer with 5+ years of experience building scalable, distributed microservices-based backend systems using Java, Spring Boot, and Apache Kafka. Specialized in event-driven architectures and real-time data pipelines handling 100M+ events/day. Strong focus on system reliability, performance optimization, and production-grade engineering. Currently exploring AI-driven backend systems.

Education

Bannari Amman Institute of Technology (BIT) | Sathy, India
Bachelor Of Engineering - Computer Science, CGPA: 8.36/10

March 2020

Work Experience

Sr. Software Development Engineer | Rocket India (formerly Mr. Cooper), Chennai, India **May 2024 – Present**

- Led large-scale Kafka cluster migration from Confluent public clusters to private clusters, successfully transitioning 200+ topics across 20+ application teams with zero data loss.
- Designed and implemented cluster linking and topic mirroring strategies, ensuring seamless data flow and zero-downtime migration.
- Developed comprehensive data lineage mapping across Kafka topics, producers, consumers, and connectors, enabling strategic migration planning and dependency resolution.
- Coordinated with cross-functional teams to execute custom migration strategies, ensuring minimal disruption across dependent systems.
- Built and scaled event-driven microservices architecture for servicing workflows with architecture designed to scale to 200+ event types.
- Introduced Flink SQL (Confluent Cloud) for real-time stream processing and transformation:
 - Built pipelines to consume from mirror topics and publish to internal topics with standardized naming conventions.
 - Automated pipeline creation using Confluent Cloud REST APIs.
 - Enabled lightweight transformations and improved data flow consistency across systems
- Contributed to data streaming solutions for Front Office Modernization, integrating 10+ data sources into centralized pipelines handling 100M+ events/day.
- Improved processing performance by 75% through batch and parallel processing enhancements.
- Enhanced pipeline reliability and reduced downtime through system optimizations.

Software Development Engineer II | Mr. Cooper, Chennai, India

May 2022 – Apr 2024

- Engineered REST APIs for efficient data production to Kafka topics, utilizing Apigee to ensure secure and scalable access across multiple teams and applications, ensuring 0 data loss for 10K+ requests/min, improving system availability to 99.9%.
- Implemented alerting jobs for real-time data production and ingestion monitoring, promptly notifying support and application teams during downtime, and generating 5+ proactive alerts per hour to ensure timely issue resolution.
- Built and standardized Kafka Connect base images to support seamless integration with diverse data sources and sinks, including relational databases, cloud storage, and data warehouses.
- Automated cloud data lifecycle policies in Azure Data Lake, saving \$150K/year.

- Played a key role in the end-to-end development of a centralized platform for Kafka topic management and access control, streamlining the management of 2000+ Kafka topics across clusters.
- Integrated the platform with cluster-level event processing metrics, billing insights, and application-specific consumer activity, enabling cost and resource monitoring. Strengthened operational visibility through stream lineage insights, identifying active and inactive resources.
- Built real-time stream processors using Kafka Streams, optimizing data enrichment and improving event handling latency by 40%.
- Created robust migration scripts to transition Kafka resources and key vaults from Azure to GCP, enabling seamless cluster migration, topic mirroring, and schema linking. Ensured 100% data integrity throughout the process, enhancing operational efficiency and reducing downtime risk.

- Developed Kafka library wrappers as Python packages and Ruby gems, simplifying integration, ensuring reliable message delivery, and enabling seamless data serialization to streamline workflows and enhance performance.
- Devised an automated functional testing tool, boosting testing capabilities and reducing manual testing effort by 50%, accelerating deployment cycles.
- Migrated applications to Azure Kubernetes Service (AKS), replacing distributed deployments and enhancing scalability and management.

Skills

- **Languages:** Java, Python, JavaScript
 - **Backend & Streaming:** Spring Boot, Apache Kafka, Kafka Streams
 - **Cloud & DevOps:** Docker, Kubernetes, Azure, GCP
 - **APIs & Tools:** REST, Apigee, Git, Shell Scripting
 - **Exploration:** AI/ML fundamentals (PyTorch, TensorFlow, NumPy)
-